Chennai Relief Control: An Explainable Disaster Relief Planner Using Rule-Based Hazard Scoring, Flood-Aware A* Routing and a Genetic Algorithm for Truck Dispatch

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***Abstract*—Chennai floods repeatedly, and relief managers must decide within hours which areas to supply first, with which trucks, and along which roads that are still open. This paper presents Chennai Relief Control, an explainable web planner that turns live rain, wind and earthquake readings into a 0–100 hazard score for each of 19 zones, converts flood hotspots into relief areas with people and supply needs, and dispatches a truck fleet using only classical artificial-intelligence techniques: fixed hazard rules, A* search on an OpenStreetMap road graph of 8,044 junctions whose roads slow or close with flood intensity, and a genetic algorithm that assigns stops to trucks. A* returns the same route cost as Dijkstra's algorithm on all 1,008 reachable routes tested. We replay six events and audit the dispatcher against nearest-depot and random-search baselines. The audit finds that the shipped genetic algorithm never improved on its starting plan, because its fitness was clamped at 0.01, and that the headline coverage counted areas with no open road as reached (66.5% reported against 43.0% by road in the 2015 floods). Removing the one-line clamp raises road coverage to 64.8% for the 2015 floods and 55.6% for Cyclone Michaung, 17.4 and 13.4 points above the best baseline.**

***Keywords*—disaster relief logistics, humanitarian supply chain, A* search, genetic algorithm, flood-aware routing, explainable AI, Chennai**

I. Introduction

Chennai is a flat coastal city crossed by tidal rivers and dotted with lakes, and it floods hard. The December 2015 floods were examined by a national performance audit of flood management and response in Chennai and its suburbs [1], and a rapid assessment by Indian research institutes reported more than 400 deaths and more than 100,000 damaged structures [2]. In December 2023, Cyclone Michaung brought about 468 mm of rain to the city in 48 hours, a total comparable to that of 2015 [3].

Planning relief for such an event combines three hard problems. First, needs must be estimated before reports arrive: which neighbourhoods are flooding, how many people are affected, and how much food, water, medicine, blankets and emergency kits they need. Second, the hazard damages the road network that the relief must use, so the fastest route in dry weather may be slow or closed. Third, a fleet with limited capacity, fuel and stock must be assigned to many stops at once. Humanitarian logistics differs from commercial logistics in its urgency, uncertainty and lack of infrastructure [4].

Decision-support tools in this setting face a second demand: planners must be able to see why a plan was chosen. Rudin argues that for high-stakes decisions, interpretable models should be preferred to explanations of black boxes [5]; explainable AI has been surveyed broadly [6]; and a recent systematic review of human–AI decision-making in disasters lists weak interpretability as a significant limitation of current systems [7].

This paper presents **Chennai Relief Control**, a web application that follows the planning chain from hazard readings to a truck dispatch with arrival times, using only classical techniques in the sense of Russell and Norvig [8]. Its contributions are:

- An end-to-end, rule-based pipeline from live weather and earthquake feeds to zone hazard scores, relief areas with quantified needs, and truck dispatch with clock-time arrivals, for 19 zones, 23 depots and 46 trucks, in which every number traces to a stated rule.
- A flood-aware road model in which hotspot intensity slows or closes nearby roads, searched by A* with an admissible and consistent travel-time heuristic whose routes match Dijkstra's on 1,008 test routes.
- A genetic-algorithm dispatcher with an eight-term fitness function, evaluated on six replayed events against nearest-depot and random-search baselines with a road-feasible coverage measure.
- An audit of the shipped system that finds and measures two faults, a fitness clamp that disabled the optimiser and a coverage headline that counted cut-off areas as reached, together with the one-line repair and its effect.

Section II reviews related work, Section III formulates the problem, Section IV describes the proposed system, Section V gives the experimental setup, Section VI reports the results and Section VII concludes.

II. Related Work

A. Humanitarian logistics and relief routing

Surveys of operations research in disaster management [9], [10], emergency logistics models [11] and humanitarian logistics [12] describe a field dominated by optimisation models. Early work formulated multi-commodity, multi-modal network flows for relief [13]. Özdamar et al. planned the dispatch of relief commodities in natural disasters [14], Barbarosoğlu et al. analysed helicopter logistics hierarchically [15], Sheu combined fuzzy clustering with optimisation to serve urgent demand [16], Balcik et al. studied last-mile distribution [17], Huang et al. compared equity, efficiency and efficacy objectives for relief routing [18], and Yi and Kumar applied ant colony optimisation [19]. These works are mostly model- and solver-centred, and do not follow live hazard readings and road conditions through to a plan that a duty officer can read.

B. Metaheuristics and learned methods for routing

Assigning trucks to stops is a vehicle routing problem [20]. Genetic algorithms are a standard heuristic for it [21], and hybrids with local search are among the best performers [22], [23]. In disaster settings, Toathom and Champrasert evolve relief routes under flood uncertainty with a dual-population genetic algorithm [24], and Insani et al. use a modified genetic algorithm to combine evacuation with supply delivery in early-stage floods [25]. Learned methods are appearing as well: Anuar et al. model multi-depot relief routing with stochastic road capacity as a Markov decision process solved by rollout [26], and Peng et al. train multi-agent deep reinforcement learning for truck–drone relief routing [27]. These methods report gains, but their decisions are hard to audit, and, as Section VI shows, a small defect in a fitness function can silently remove all optimisation.

C. Flooded roads and graph search

Flooding degrades road transport gradually before it stops it: Pregnolato et al. relate vehicle speed to flood depth with a depth–disruption function [28]. Route search on road graphs uses Dijkstra's algorithm [29] or A*, which adds an admissible heuristic to guide the search [30]. Road graphs can be extracted from OpenStreetMap [31]. We adopt the idea of graded slowdown and add outright closure for the most severe hotspots (Section IV-D).

D. Positioning

Table I positions this work. To our knowledge, none of the surveyed works combines rule-based hazard scoring, flood-aware A* routing, genetic dispatch and plain-language reasons in one deployed tool, and none reports an audit of its optimiser against baselines on replayed events.

TABLE I
COMPARISON WITH RELATED WORK

Work	Focus	Technique	Learned model
[14]	Relief dispatch	Mathematical programming	No
[16]	Urgent relief demand	Fuzzy clustering + optimisation	No
[19]	Relief distribution	Ant colony optimisation	No
[24]	Flood relief routes	Dual-population GA	No
[25]	Early-stage flood relief	Modified GA	No
[26]	Relief, uncertain roads	MDP + rollout (RL)	Yes
[27]	Truck–drone relief	Multi-agent deep RL	Yes
[28]	Flood effect on roads	Depth–disruption function	Empirical
Ours	**Chennai relief dispatch**	**Rules + A* + GA**	**No**

III. Problem Formulation

A. Study area and events

The study area is Chennai and its suburbs: the 15 zones of the city corporation and four neighbouring municipalities (Avadi, Pallavaram, Chrompet and Tambaram), 19 zones in all, covering 712 kmZ with a modelled population of 6.48 million. Zone and ward shapes come from Datameet and Transparent Chennai. We replay six events: four past events (the 2015 floods, Cyclone Vardah 2016, Cyclone Michaung 2023 and Cyclone Fengal 2024) from rain and wind figures held in the application, and two drills (a severe cyclone and an M6.6 earthquake).

B. Entities

A plan serves a set of relief areas (at most 60) from 23 depots with 46 trucks. Each area has a number of people affected, a severity class (Critical, High, Medium or Low, weighted 4, 2.5, 1.5 and 1) and a demand for five item types: food, water, medicine, blankets and emergency kits. An area whose demand exceeds 5,000 units is split into several deliveries, so the 57 areas of the 2015 replay become 69 stops. A truck carries 1,200–6,000 units (median 2,500), holds 60–200 L of fuel, burns 0.12–0.30 L/km and travels at 32–40 km/h. Each truck makes one trip, so the fleet can deliver at most 166,700 units in total. Depots, stock, fleet and the list of 57 flood hotspots are representative placeholders, and populations are modelled estimates.

C. Objective and measures

A plan gives each truck an ordered list of stops; each stop appears in exactly one list; every leg follows the A* route over the road graph in its flooded state; and a leg is *cut off* if A* finds no open road. The planner maximises the severity-weighted relief delivered while penalising distance, overload, fuel overuse, cut-off legs and imbalance (Eq. 8). We report two coverage measures. *Reported coverage* is the share of requested relief units that the application shows as delivered. *Road coverage* counts a stop only if every leg from the depot up to and including that stop found an open

road, so it is the share of requested units that a truck could actually deliver. We also report the number of Critical areas reached, the number of cut-off areas, total distance, and the mean arrival time of the reached areas weighted by people and severity.

IV. Proposed System

A. Architecture

Chennai Relief Control has four layers (Fig. 1). A FastAPI service in Python refreshes the feeds every 15 minutes from Open-Meteo [32] (rain and wind gusts at seven points, waves, and a river-discharge index) and the USGS earthquake catalogue [33] (events within 600 km in the last three days), scores the zones, derives relief areas, updates the state of the roads and plans the dispatch. A React and TypeScript client shows the map, the dispatch plan, the resources and the reports. State is kept in SQLite and plans export as PDF run sheets.

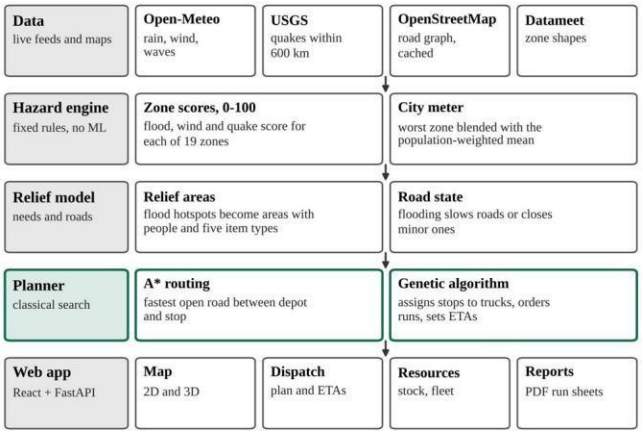


Fig. 1. Layered architecture of Chennai Relief Control. The planner (highlighted) is the only part that searches; the hazard engine is fixed rules.

B. Hazard scoring

Each zone z receives flood, wind and quake scores from 0 to 100. The flood score is

$$F_z = 100\,\rho\,\chi_z\,(1 + 0.12\,r\,L_z) \tag{1}$$

where $\rho = \max(f_{24}(R_{24}), f_{72}(R_{72}))$ is a piecewise-linear rain factor of the 24-hour and 72-hour totals, anchored on the IMD classes heavy (64.5 mm), very heavy (115.6 mm) and extremely heavy (204.5 mm) [34]; L_z in [0, 1] is how low-lying the zone is; r is the river index; and $\chi_z = 0.40 + 0.35\,L_z + 0.25\,H_z$ is susceptibility, with $H_z = \min(1, \Sigma\, s_h / 9)$ summing the severity s_h (1–3) of the zone's flood hotspots. The wind score applies a piecewise-linear gust factor with knots at 62, 89, 118 and 168 km/h, approximately the lower bounds of the IMD cyclonic-storm classes [34], scaled by 0.60 (inland) or 1.00 (coastal), together with a storm-surge term. The quake score is 100(MMI − 3)/5, clipped to 0–100, from a magnitude–distance intensity estimate with soil amplification.

The zone score combines the largest two hazards, and the city meter blends the worst zone with the population-weighted mean:

$$S_z = \min(100,\, m_1 + 0.30\,m_2) \tag{2}$$

$$M = \min(100,\, 0.55\,\max_z S_z + 0.45\,\Sigma\, p_z\,S_z / \Sigma\, p_z) \tag{3}$$

where $m_1 \geq m_2$ are the two largest of the zone's flood, wind and quake scores and p_z is its population. The meter is Elevated from 25, High from 50 and Severe from 75. Because M is non-decreasing in every score, it cannot fall when a hazard rises (verified in Section VI-A). Each score is shown with its drivers, for example "rain 180 mm/24 h, low-lying 0.8, 4 flood hotspots".

C. Relief areas and needs

A flood hotspot h in zone z with severity s_h and exposed population P_h has local intensity

$$I_h = \min(1, (F_z / 100)(0.55 + 0.45\,s_h / 3)) \tag{4}$$

and becomes a relief area when $I_h \geq 0.30$. The people affected follow from the intensity:

$$n_h = \max(60, \text{round}(0.18\,P_h\,q_h^{\,1.15})) \tag{5}$$

where $q_h = \min(1, (I_h - 0.25) / 0.75)$. Severity is Critical for $I_h \geq 0.78$, High for 0.58, Medium for 0.40 and Low below. Needs for the five item types are fixed per-person quantities that depend on the hazard type. Zone-level storm and quake areas are added when the wind or quake score reaches 40. Every area stores its reason, for example "flood intensity 78/100 (zone flood score 82, hotspot severity 3/3, rain 190 mm/24 h)".

D. Flood-aware routing

Roads come from OpenStreetMap and form a graph of 8,044 junctions and 10,474 segments in 153 connected components, of which the largest holds 7,716 junctions (95.9%). Segment classes have free-flow truck speeds v_c of 55 (motorway), 42 (trunk), 34 (primary), 28 (secondary), 24 (tertiary), 18 (residential) and 15 km/h (link). Around each hotspot of radius a_h, a segment at distance d is slowed by

$$\varphi = \max(0.18,\, 1 - 0.85\,k\,(1 - 0.45\,d / a_h)) \tag{6}$$

with $k = \min(1, (I_h - 0.2) / 0.7)$, so even the worst slowdown leaves 18% of free-flow speed. A minor road is closed outright when $I_h \geq 0.82$, $s_h = 3$ and $d \leq 0.65\,a_h$. The travel time of a segment of length ℓ is

$$t = \ell / (v_c\,\varphi\,\kappa) \tag{7}$$

where κ scales the truck's speed against a 36 km/h reference. A* [30] orders junctions by $g(u) + h(u)$, with $h(u)$ the great-circle distance to the goal divided by the highest speed in the network. Because every segment is at least as long as the great-circle distance between its ends and cannot be driven faster than that speed, h is admissible and consistent, so A* returns a least-time route (Algorithm 1).

Algorithm 1: Flood-aware A* search

```
function AStar(s, g)
    open ← {s}; cost[s] ← 0
    while open ≠ ∅ do
        u ← junction in open with least cost[u] + h(u)
        if u = g then return path to g
        for each segment (u, v) with φ > 0 do
            c ← cost[u] + ℓ / (v_c φ κ)
            if c < cost[v] then cost[v] ← c; add v to open
    return NOT FOUND   ▷ leg is cut off
```

E. Dispatch by genetic algorithm

A chromosome maps each truck to an ordered list of stops. The initial population holds two seeded plans, a severity-ranked round robin (seed 1) and a nearest-depot assignment to the least-loaded truck (seed 2), and random plans. Each generation keeps the two best plans and fills the rest with children: parents are chosen by tournament of four [35], recombined with order crossover (OX1) [36] with probability 0.80, and mutated with probability 0.05 by moving a stop, reassigning it to another truck, or reversing part of a route (Fig. 2). The balanced setting uses 100 plans and 250 generations; Fast uses 60 and 100, and Best 160 and 600. The fitness of a plan is

$$F = 3P + 2000c - 1.5D - 80O - 100E - 500U - 10\sigma - 300|N| \quad (8)$$

where P is the severity-weighted units delivered, c the share of areas that appear in some route, D the total distance in km, O the units loaded beyond truck capacity, E the litres beyond a tank, U the number of cut-off legs, σ the standard deviation of stops per truck and N the unserved areas. A cut-off leg also adds a stand-in distance of 9,999 km. Table II lists the terms. Three of the eight terms (delivered priority, coverage and unserved areas) take the same value for every plan, because every stop is in some route.

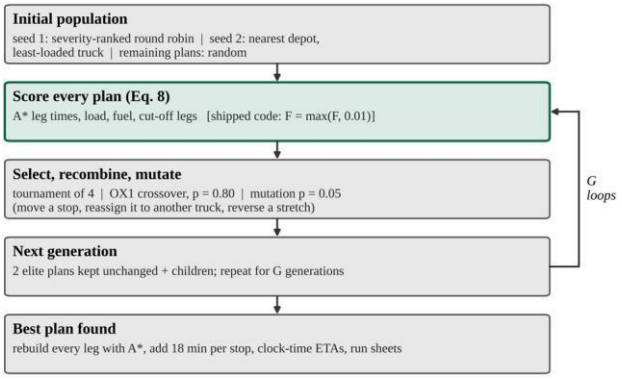


Fig. 2. Cycle of the genetic algorithm. The scoring step is where the shipped code clamps fitness at 0.01.

TABLE II
FITNESS TERMS OF THE DISPATCH PLAN

Term	Weight	Unit	Varies by plan
Priority-weighted delivery	+3	unit × weight	No
Area coverage	+2000	fraction	No
Unserved areas	−300	per area	No
Distance	−1.5	per km	Yes
Truck overload	−80	per unit	Yes
Fuel beyond tank	−100	per litre	Yes
Cut-off leg	−500	per leg	Yes
Fleet imbalance	−10	× σ of stops	Yes

Algorithm 2: Genetic dispatch

```
Pop ← 2 seeded plans + random plans
for gen = 1 to G do
    score every plan with Eq. (8) using A* leg times
    Next ← the 2 best plans (elitism)
    while |Next| < |Pop| do
        pick two parents by tournament of 4
        child ← OX1 with prob. 0.80, else a copy
        mutate child with prob. 0.05; add to Next
    Pop ← Next
rebuild legs of the best plan with A*; add 18 min per stop
return best plan, ETAs in IST
```

F. Application and explanations

The Dispatch page (Fig. 3) opens with one sentence on the plan, four headline numbers (first delivery, Critical areas, people and relief delivered), a warning when stock is short, a list of areas that are cut off, and a replay of the trucks on a 2D or 3D map. Arrival times are clock times in IST that add 18 minutes of unloading per stop. Run sheets export as a PDF, and the Reports page states the rules in plain words.

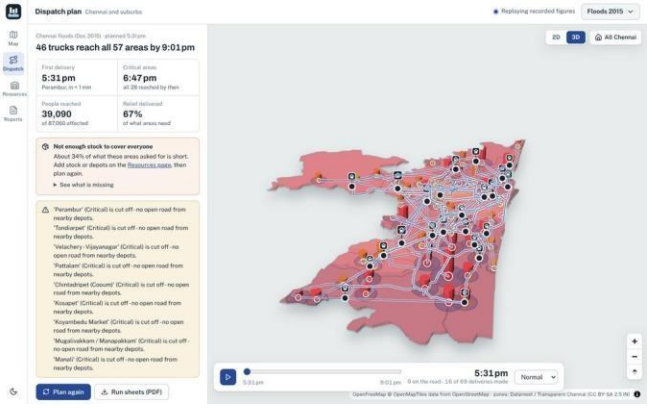


Fig. 3. Dispatch page for the 2015 floods replay. The headline reads "46 trucks reach all 57 areas by 9:01 pm", while the warnings below it list areas with no open road.

V. IMPLEMENTATION AND EXPERIMENTAL SETUP

The back end is Python with FastAPI and SQLite (4,736 lines in 21 files) and the front end is React 18 with TypeScript (4,166 lines in 28 files). The 14 shipped tests pass in 1.8 s. The experiments call the real API: for each event, the scenario is loaded and a plan is generated with 100 plans, 250 generations and up to 60 areas. Table III gives the model outputs for the six events.

TABLE III
REPLAYED EVENTS AND MODEL OUTPUTS

Event	Meter	Areas (Crit.)	People	Items	Roads slow / closed
Floods 2015	94.1	57 (28)	87,060	191,179	1,893 / 219
Michaung 2023	96.4	60 (25)	118,660	261,489	1,903 / 209
Fengal 2024	75.4	60 (0)	55,090	121,448	2,112 / 0
Vardah 2016	90.1	60 (7)	146,560	325,109	1,537 / 0
Cyclone drill	94.7	60 (13)	198,370	439,175	2,112 / 0
Quake drill	63.7	19 (0)	138,200	353,807	0 / 0

Four strategies are compared. *Shipped* is the genetic algorithm as released. *Nearest depot* is seed 2, a one-pass rule with no search. *Random search* scores 25,000 random

plans, the same number of evaluations as the genetic algorithm, with the clamp-free fitness and keeps the best. *Clamp removed* is the genetic algorithm with the one line `fitness = max(raw, 0.01)` changed to `fitness = raw`. Shipped, random search and clamp removed use three seeds, and nearest depot is deterministic. We report the mean and the minimum–maximum range. Road coverage and the other measures are computed from the legs of each returned plan (Section III-C).

VI. Results and Discussion

A. Correctness and speed

Routing. We sampled 400 random junction pairs for each of three events, 1,200 in all; 1,008 had a route. A* and Dijkstra returned the same travel time on all 1,008. A* expanded about 30% fewer junctions but ran 20–40% slower in pure Python, because of heap and heuristic overhead, so we claim no speed advantage. Flooding changes routes substantially. Of 300 depot–area pairs per event in the 2015 replay, 54 (18%) had no open road; of the 246 that did, 15% had a dry-weather route crossing a closed road, 67% had a different best route, and the mean travel time was 19.9% longer. For Michaung the figures are 22%, 13%, 65% and 17.2%, and for the cyclone drill 3%, 0%, 49% and 11.7%.

Hazard model. Sweeps of rain (16 steps), wind (19 steps) and quake magnitude (8 steps) showed that the city meter never fell as a hazard rose, and zone scoring takes 0.16 ms. A first plan for a new event takes 41–51 s while the A* cache fills, and later plans 7–12 s. The interface showed no horizontal overflow at six widths on four pages.

B. Finding 1: the fitness clamp disables the optimiser

Every plan in the 2015 replay has about 35 cut-off legs, each carrying the 9,999 km stand-in, so raw fitness is about −10 to −12 million for all of them (Table IV). The shipped code applies `fitness = max(raw, 0.01)` (line 318 of the genetic module), which maps every plan to the floor of 0.01. Tournaments then cannot prefer any plan, the best-so-far record never changes, and the plan returned is the first seed, the severity-ranked round robin. The convergence log confirms it: the best fitness is 0.01 in every generation and the best distance is the same at generation 0 and generation 225 in all six events (grey line in Fig. 5). Removing the clamp changes one line and the 14 tests still pass.

TABLE IV
Starting Plans and Result, Floods 2015

Plan	Distance (km)	Cut-off legs	Overload (units)	Fitness
Seed 1: severity round robin	351,433	35	63,976	0.01
Seed 2: nearest depot	350,420	35	52,315	0.01
Best random plan	341,471	34	80,787	0.01
Shipped result	**351,433**	**35**	**63,976**	**0.01**

Distance includes 9,999 km for each cut-off leg. The shipped result equals seed 1.

C. Finding 2: reported coverage counts cut-off areas as reached

When A* finds no route, the plan builder records the leg as 0 km and 0 h, so the stop looks reached at once. In the 2015 replay the application reported 66.5% of relief delivered and 39,090 of 87,060 people reached (Fig. 3). Counting only stops whose legs all found an open road gives 43.0% and 26,160 people. Of the 57 areas, 24 are cut off, and 19 of 28 Critical areas are reached by road, while the screen says all 28 are reached by 6:47 pm. The gap between reported and road coverage is 23.5 points in the 2015 floods, 20.7 for Michaung, 4.8 for Fengal, 7.4 for Vardah, 6.2 for the cyclone drill and 6.5 for the earthquake drill. Closed roads alone make some areas unreachable: 399 of 1,587 depot–area pairs (25%) have no open route in 2015.

D. Optimiser against baselines

Table V and Fig. 4 give road coverage. The shipped optimiser is never the best strategy. It trails random search on all six events, by 1.7 to 4.8 points, and trails the nearest-depot rule on Fengal by 5.9 points. With the clamp removed, the genetic algorithm gives the highest road coverage on every event: 64.8% against 47.4% for the 2015 floods and 55.6% against 42.2% for Michaung, gains of 17.4 and 13.4 points over the best baseline, with run-to-run ranges that do not overlap those of random search. On the cyclone drill the ranges of the two methods overlap (34.7–36.1% and 35.4–36.0%), so we count it a tie.

TABLE V
Road Coverage (%) by Strategy

Event	Shipped	Nearest	Random	Clamp off	Gain	Limit
Floods 2015	43.0	43.4	47.4	64.8	+17.4	87.2
Michaung 2023	37.4	33.5	42.2	55.6	+13.4	63.8
Fengal 2024	79.1	85.0	81.4	94.4	+9.4	100
Vardah 2016	42.3	40.8	44.0	46.5	+2.5	51.3
Cyclone drill	31.5	30.9	35.2	35.7	+0.5	38.0
Quake drill	39.4	33.3	41.1	44.4	+3.3	47.1

Means of three runs (nearest depot is deterministic). Gain is clamp removed minus the best baseline. Limit is the one-trip fleet limit, 166,700 units divided by the items requested.

The size of the gain follows the headroom. Because every truck makes one trip, road coverage cannot exceed 166,700 units divided by the items requested, which is 87.2% for the 2015 floods but only 38.0% for the cyclone drill (the “Limit” column). The repaired algorithm reaches 74% of this limit on the 2015 floods and 87–94% on the other five events. Where the best baseline is far below the limit, as in the 2015 floods (39.8 points of headroom), the gain is large; where it is close, as in the cyclone drill (2.8 points), little can be gained.

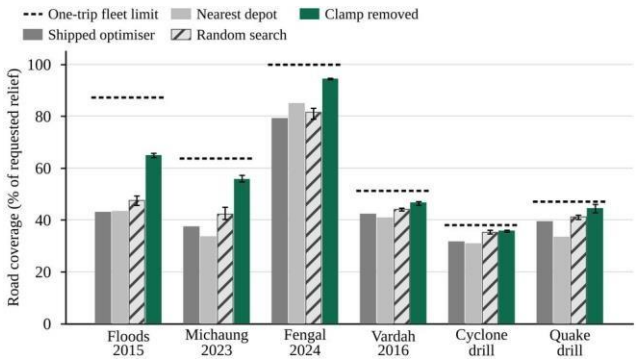


Fig. 4. Road coverage of each strategy. Whiskers show the minimum–maximum of three runs; dashed lines mark the one-trip fleet limit.

E. Detail for the 2015 floods

TABLE VI
STRATEGIES ON THE 2015 FLOODS

Strategy	Rep. %	Road %	Crit.	Cut off	Dist. km	Arr. h
Shipped	66.5	43.0	19	24	1,468	0.84
Nearest depot	72.6	43.4	18	17	454	0.49
Random search	64.4	47.4	17.3	13.7	1,589	1.19
Clamp removed	72.8	64.8	19	9	1,549	0.93
+ free cut-off	82.9	51.7	19	14.7	1,003	0.65

Rep. is reported coverage. Crit. is Critical areas reached (of 28). Cut off is areas with no open road (of 57). Arr. is the mean arrival of reached areas weighted by people and severity, so a plan that reaches fewer areas can look faster.

Clamp removed reaches as many Critical areas as the shipped plan (19 of 28) but cuts off only 9 areas instead of 24, for 81 km more travel and a weighted arrival of 0.93 h against 0.84 h. Nearest depot is the shortest (454 km, 0.49 h) but leaves 17 areas cut off. The last row tests the obvious repair of the reporting fault, making cut-off legs cost nothing. It backfires: the reported figure rises to 82.9%, but road coverage falls to 51.7% and 14.7 areas stay cut off, consistent with removing the 9,999 km stand-in so that only the 500-point leg penalty remains. Reporting must therefore be corrected without relaxing the penalty. Fig. 5 shows the clamp-free search cutting the cut-off legs of its best plan from 30–35 to 22 (2015) and 19 (Michaung), while the shipped plan stays at 35.

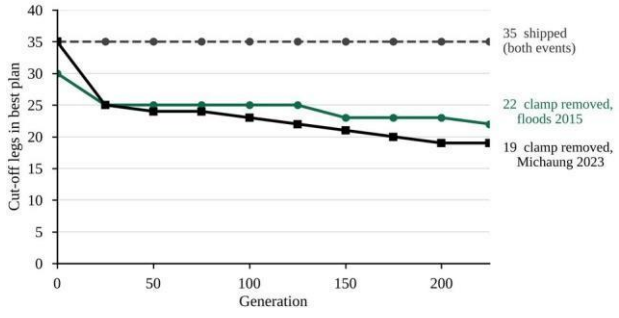


Fig. 5. Cut-off legs in the best plan by generation, first of three runs, for the shipped optimiser and the clamp-free search.

F. Search effort

With the clamp removed, more search helps, with diminishing returns. Mean road coverage over three runs is 57.8%, 47.1% and 90.3% with Fast (60 plans, 100 generations) for the 2015 floods, Michaung and Fengal; 64.3%, 53.0% and 94.0% with Balanced (100, 250); and 66.5%, 56.1% and 94.7% with Best (160, 600). Once the A* cache is warm, compute time is about 6–14 s for Fast, 8–11 s for Balanced and 25–34 s for Best. These runs used other seeds than Table V, so the Balanced means differ slightly.

G. Limitations and threats to validity

Depots, stock, fleet and hotspots are placeholders and populations are modelled, so the coverage figures describe this model and not the real fleet. The four past events are replays of rain and wind figures, not reconstructions of the true road state, and the closure rule has not been calibrated against observed closures; depth–disruption evidence [28] suggests that real thresholds depend on water depth and vehicle type. The repaired algorithm and random search both optimise the road-feasibility signal that road coverage measures, which favours methods that share the objective. Each cell has only three runs, and the ranges are minimum–maximum, not confidence intervals. Trucks make one trip, areas are capped at 60 and large areas are split at 5,000 units. Earthquake damage to roads is not modelled. The repair was measured on a patched copy and is not yet in the released application, and no user study has tested the explanations.

VII. CONCLUSION AND FUTURE WORK

Chennai Relief Control shows that classical AI can carry live weather and earthquake readings through to a readable truck dispatch: fixed rules score 19 zones and create relief areas, flood-aware A* finds the fastest open roads, and a genetic algorithm assigns stops to trucks, with a reason attached to each number. Auditing the system against baselines on six replayed events exposed two faults that no test had caught. A fitness clamp made the shipped optimiser return its starting plan, and a coverage headline counted cut-off areas as reached. Removing the clamp raises road coverage to 64.8% for the 2015 floods and 55.6% for Michaung, and the gain follows the headroom left under the one-trip fleet limit.

Future work includes applying the repair and showing reported and road coverage side by side on the Dispatch page, allowing several trips per truck, calibrating the closure thresholds against observed flood depths and road closures, comparing against hybrid genetic search [23], modelling road damage from earthquakes, and a user study with relief planners on whether the explanations speed up and improve decisions.

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